Tutorial 5 (Friday)

February 6, 2025 11:30 AM

Problem #1: Find a power series representation of the following function and determine the radius of convergence.

$$f(x) = \frac{x}{8+x^4}$$

$f(x) = x \cdot \frac{1}{8\left(1+\frac{x^4}{8}\right)} = x \cdot \frac{1}{8\left(1-\left(\frac{-x^4}{8}\right)\right)} = \frac{x}{8}\sum_{n=0}^{\infty}\left(\frac{-x^4}{8}\right)^n$

$= \sum_{n=0}^{\infty}\frac{(-1)^n}{8^{n+1}}(x^{4n+1})$ and $\left|\frac{-x^4}{8}\right| < 1 \Rightarrow |x^4| < 8$

$|x| < \sqrt[4]{8}$

radius of convergence.

Problem #2: Evaluate the following integral as a power series.

$$\int \ln(1+x^7)\,dx$$

Note: $f(x) = \ln(1+x^7)$

$f'(x) = \frac{1}{1+x^7}(7x^6) = 7x^6\frac{1}{1-(-x^7)} = 7x^6\sum_{n=0}^{\infty}(-1)^n(x^{7n})$

$= \sum_{n=0}^{\infty}7(-1)^n x^{7n+6} = 7(-1)^0x^{7(0)+6} + 7(-1)^1x^{7(1)+6} + 7(-1)^2x^{7(2)+6} + \dots$

$= 7x^6 - 7x^{13} + 7x^{20} - \dots$

So, $\ln(1+x^7) = \int 7x^6 - 7x^{13} + 7x^{20} - \dots)\,dx$

$= \frac{7x^7}{7} - \frac{7x^{14}}{14} + \frac{7x^{21}}{21} - \dots + C$

$= x^7 - \frac{x^{14}}{2} + \frac{x^{21}}{3} - \dots + C$

So, $\int \ln(1+x^7)\,dx = \int x^7 - \frac{x^{14}}{2} + \frac{x^{21}}{3} - \dots + C\,dx$ (Antiderivative)

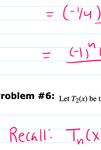
$= \frac{x^8}{8} - \frac{x^{15}}{15\cdot 2} + \frac{x^{22}}{22\cdot 3} - \dots + C'$

$= \sum_{n=0}^{\infty}\frac{(-1)^n x^{8+7n}}{(n+1)(8+7n)}$

Problem #3: Find the sum of the following series.

$\sum_{n=0}^{\infty}\frac{(-1)^n\pi^{2n}}{6^{2n}(2n)!}$ looks like $\cos(x) = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!}$

$= \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\left(\frac{\pi}{6}\right)^{2n} = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$



Problem #4: Find the Taylor series for $f(x) = -\frac{1}{x}$ centered at $a = 2$.

$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$

Note: $f'(x) = x^{-2}$ $f^{(2)}(x) = -2x^{-3}$ $f^{(3)}(x) = 6x^{-4}$

pattern: $f^{(n)}(x) = (-1)^{n-1}x^{-n-1}n!$ $f^{(0)} = -x^{-1}$

So: $f(x) = \sum_{n=0}^{\infty}\frac{(-1)^{n+1}n!}{2^{n+1}}\frac{(x-2)^n}{n!}$

$= \sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{2^{n+1}}(x-2)^n$ Sometimes you try $(-1)^{n+1}$ instead.

Problem #5: Find a simplified expression for $\binom{-1/4}{n}$.

Recall: $\binom{u}{k} = \frac{u(u-1)(u-2)\cdots(u-k+1)}{k!}$

$\binom{-1/4}{n} = \frac{(-1/4)(-1/4-1)(-1/4-2)\cdots(-1/4-n+1)}{n!}$

$= \frac{(-1/4)(-5/4)(-9/4)\cdots(3/4-n)}{n!}$ $\frac{3-4n}{4}$

$= \frac{(-1)^n(1\cdot 5\cdot 9\cdots(4n-3))}{4^n(n!)}$

Problem #6: Let $T_2(x)$ be the 2nd degree Taylor polynomial for the function $f(x) = 9 + xe^{-3x}$ centered at $a = 0$. Find $T_2(1)$.

Recall: $T_n(x) = \sum_{i=0}^{n}\frac{f^{(i)}(a)}{i!}(x-a)^i$

So, $T_2(x) = \sum_{i=0}^{2}\frac{f^{(i)}(0)}{i!}(x)^i$

$= \frac{f^{(0)}(0)}{0!}x^0 + \frac{f^{(1)}(0)}{1!}x^1 + \frac{f^{(2)}(0)}{2!}x^2$

$f^{(0)}(x) = 9 + xe^{-3x}$

$f^{(1)}(x) = e^{-3x} + x(-3)e^{-3x}$

$= e^{-3x}(1-3x)$

$f^{(2)}(x) = (-3)e^{-3x}(1-3x) + e^{-3x}(-3)$

$= (-3)e^{-3x}[2-3x]$

$f^{(0)}(0) = 9$

$f^{(1)}(0) = 1$

$f^{(2)}(0) = -3\cdot 1\cdot 2 = -6$

$T_2(x) = 9 + x + \frac{(-6)}{2!}x^2 = 9 + x - 3x^2$

$T_2(1) = 9 + 1 - 3 = 7$ (wow...)

Problem #7: Let $f(x) = \frac{1}{x}$, $0.3 \le x \le 1.7$. Suppose that we approximate $f(x)$ by the 4th degree Taylor polynomial $T_4(x)$ centered at $a = 1$. Taylor's inequality gives an estimate for the error involved in this approximation.

Find the smallest possible value of the constant M referred to in Taylor's Inequality.

if $|f^{(n+1)}(x)| \le M$

$\Rightarrow |R_n(x)| = |f(x) - T_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$

$T_4(x)$ in this case.

WTFind: bound on $f^{(5)}(x)$.

$f^{(0)}(x) = x^{-1}$ $f^{(1)}(x) = -x^{-2}$ $f^{(2)}(x) = 2x^{-3}$

$f^{(3)}(x) = -3!x^{-4}$ $f^{(4)}(x) = 4!x^{-5}$ $f^{(5)}(x) = -5!x^{-6}$

So, $|f^{(5)}(x)| = \frac{5!}{x^6} \le \boxed{\frac{5!}{(3/10)^6}}$ → ok for Childsmath.

Note: $0.3 \le x \le 0.7$

$\frac{1}{3/10}$

Problem #8: The curve $y = \sqrt{9-x^2}$, $-2 \le x \le 1$, is rotated about the x-axis. Find the area of the resulting surface.

Surface area rotating about the x-axis:

$S = \int_a^b 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx$

Here $a = -2$, $b = 1$ $f(x) = \sqrt{9-x^2}$

$f'(x) = \left(\frac{1}{2}\right)(9-x^2)^{1/2}(-2x) = \frac{-x}{\sqrt{9-x^2}}$

$(f'(x))^2 = \frac{x^2}{(9-x^2)}$

So, $\int_{-2}^{1} 2\pi\sqrt{9-x^2}\sqrt{1+\frac{x^2}{9-x^2}}\,dx$ $\frac{9-x^2+x^2}{9-x^2} = 9$

$= \int_{-2}^{1} 2\pi\sqrt{9-x^2}\sqrt{\frac{9}{9-x^2}}\,dx$

$= \int_{-2}^{1} 6\pi\,dx = 6\pi x + c\Big|_{-2}^{1} = 6\pi(1-(-2)) = 18\pi$

Problem #9: The below curve is rotated about the y-axis. Find the area of the resulting surface.

$y = \frac{x^4}{16} + \frac{1}{2x^2}$, $1 \le x \le 2$

Surface area rotating about the y-axis:

$S = \int_a^b 2\pi x\,ds$ $ds = \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$

or: $ds = \sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$

So, $y = \frac{x^4}{16} + \frac{1}{2x^2}$ $dy = \left(\frac{4x^3}{16} + \frac{1}{2}(-2)x^{-3}\right)dx$

$\Rightarrow \frac{dy}{dx} = \left(\frac{x^3}{4} - \frac{1}{x^3}\right) = \frac{x^6-4}{4x^3}$

$\Rightarrow \left(\frac{dy}{dx}\right)^2 = \left(\frac{x^6-4}{4x^3}\right)^2 = \frac{x^{12}-8x^6+16}{16x^6}$

$\Rightarrow \sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{\frac{16x^6+x^{12}-8x^6+16}{16x^6}} = \frac{\sqrt{x^{12}+8x^6+16}}{4x^3}$

$= \frac{\sqrt{(x^6+4)^2}}{4x^3} = \frac{x^6+4}{4x^3}$ do i like calc anymore..?

$\int_1^2 2\pi x\left(\frac{x^6+4}{4x^3}\right)dx = \int_1^2 2\pi\left(\frac{x^6+4}{4x^2}\right)dx = \int_1^2 2\pi\left(\frac{x^4}{4}+\frac{1}{x^2}\right)dx$

all you need for childsmath.